

Canadian Mortgage Renewal Risk Project

Step-by-Step Guide — For Me to Re-Learn / Re-Run This Later

This is my own walkthrough notes, written so future-me (or anyone else) can understand exactly what each part of the project does and why, without re-reading every line of code.

1 Understand the folder layout

data/ holds raw inputs, sql/ holds the schema and queries, python/ holds the three scripts that do all the work, db/ holds the resulting SQLite database, charts/ holds the PNG outputs, and reports/ holds these PDFs. Everything downstream reads from data/ and writes into db/ and charts/, so those two folders are safe to delete and regenerate at any time.

2 (Optional) Refresh live data

Run `fetch_data.py`. It hits the Bank of Canada's free Valet API (no key needed) for two series: the target overnight rate (V39079) and the 5-year Government of Canada bond yield (BD.CDN.5YR.DQ.YLD), and saves them as CSVs in data/. If this fails (e.g. no internet in a sandboxed environment), the project still works using the pre-loaded, real, verified rate history in `data/boc_rate_history_seed.csv`.

```
python python/fetch_data.py
```

Note: Safe to skip. Everything else works from the seed CSV.

3 Build the database

Run `load_to_db.py`. It does three things in order: (1) runs `sql/schema.sql` to create three empty tables, (2) loads the rate history CSV (live if present, otherwise the seed file) into the `rate_history` table, (3) inserts five illustrative mortgage scenarios into `mortgage_scenario`. This is the step that actually creates `db/mortgage_risk.db`.

```
python python/load_to_db.py
```

4 Understand the schema

`rate_history` is a fact table: one row per Bank of Canada rate decision date, with the rate level, the change in basis points, and whether it was a hike/cut/hold. `mortgage_scenario` is a small reference table of sample borrowers I made up (origination date, amount, rate, term). `renewal_shock` is empty until `analysis.py` fills it in — it's the derived output table joining the two, storing the calculated payment impact per borrower.

5 Run the analysis

Run `analysis.py`. For each `mortgage_scenario` row, it: figures out the renewal date (origination date + `term_years`), applies the standard Canadian mortgage payment formula (payments are monthly but the rate compounds semi-annually by law) to get the original payment, estimates today's renewal rate from the `ASSUMED_TODAY_FIXED_RATE` constant, recalculates the payment on the remaining balance/amortization, and writes the dollar and percentage change into `renewal_shock`. It then saves two PNG charts into `charts/`.

```
python python/analysis.py
```

Note: Edit `ASSUMED_TODAY_FIXED_RATE` at the top of `analysis.py` to reflect the current market rate whenever you re-run this in the future.

6 Explore the SQL myself

Open `sql/queries.sql` — each of the 8 queries is commented with what it's for. The ones worth re-studying are #3 and #4 because they use window functions (a rolling `AVG()` `OVER (...)` and a date-ranged `SUM()` `OVER (...)`), which is exactly the kind of SQL asked about in analyst interviews. Query #5 finds cycle turning points using `LAG()` to detect when `decision_type` changes from the previous row.

```
sqlite3 db/mortgage_risk.db < sql/queries.sql
```

Quick reference: the mortgage payment formula

Canadian mortgages compound semi-annually but pay monthly, which is different from the US convention — this trips a lot of people up, so it's worth remembering explicitly:

```
i_semi = annual_rate / 2    →    i_monthly = (1 + i_semi)^(2/12) - 1    →    payment = P ×  
[ i_monthly × (1+i_monthly)^n ] / [ (1+i_monthly)^n - 1 ], where n = amortization years  
× 12.
```

If I want to extend this project later

- Add StatCan CPI/inflation data as a second series to correlate against renewal risk.
- Replace the flat `ASSUMED_TODAY_FIXED_RATE` with the live 5yr bond yield + an assumed lender spread.
- Add more `mortgage_scenario` rows, or read them from a CSV instead of hardcoding.
- Wrap the whole thing in a small Streamlit app so it's an interactive, linkable dashboard.
- Push to GitHub with this same README and link it directly on my resume.

See `Project_Overview.pdf` for the goal, motivation, and findings write-up.